

Design Document

Jumpx (Main Class):

- Jumpx is the public main class that extends JFrame and implements the ActionListener and KeyListener interfaces.
- Set up the basics for the frame
- Insert the images to represent the dino, star, moon, and the ground.
- Control the state of game, including start the game, end the game, restart the game.
- Generate the subclasses, including obstacles, stars, moons, and highscorelist.
- Load the sound clip into the game so that it will play when the dino jumps.
- Uses timers to control the generation of obstacles, stars and moons, update the score and the shows the highest score, and trigger the game loop to continue play the game.
- In the CheckCollision method, if statement is used to detect whether the dino hits the obstacle or a star or a moon, and generates different consequences.
- Use paint method to draw the images that is inserted and the obstacle. The random method is used to draw the obstacles in different sizes. It also repaint after restarting the game.
- Create an option panel to show the current score and highest score, detect the user input, and restart the game.

Obstacle:

- Represents an obstacle in the game, which is a barrier that a dino needs to jump over.
- Initialize the size and speed of the obstacle.
- The move method is used for updating the obstacle's position.
- The speed method is responsible for changing its speed.
- Getters are used to protect the properties.

Star (Coin Icon) :

- Insert a coin image to represent the star.
- Initialize the properties of the star, including size and speed.
- Use constructor to set up the basic status of the star.
- The move method is used for updating the speed of the star.
- Getters are used to protect the properties.

Moon (Sunflower Icon) :

- Insert a sunflower image to represent the star.
- Initialize the properties such as mx (x-coordinate), my (y-coordinate), and md (diameter). The move method is responsible for updating the moon's position.
- Use a constructor to set up the basic status of moon.

HighScoreList:

- Represents a list of high scores.

- Use arraylist to store all the score during one round.
- Add new scores into the arraylist.
- Provides methods to retrieve the highest score and add a new score to the list.

Relationships:

- “Jumpx” has an instance of “HighScoreList” to manage high scores.
- “Jumpx” contains lists of “Obstacle” and “Star” instances to manage the game elements. The “Obstacle” and “Star” classes encapsulate the properties and behavior of obstacles and stars, respectively.
- The “Jumpx” class interacts with the “Timer” class to control the timing of obstacle and star generation, score updates, and the game loop.
- “Jumpx” communicates with the “JOptionPane” class to display messages and receive user input.
- The game elements (obstacles, stars, and player character) are painted on the JFrame using the paint method.